

SRS Part 2

SRS Part 2 — External Interface Requirements, Analysis Models, System Features (5.1-5.7)

Role-Specific AI Voice Assistant Framework for Organizations and Entities — Jackfruit Phase-1, Team NexGen (Team ID 43)

3. External Interface Requirements

3.1 User Interfaces

The Role-Specific AI Voice Assistant Framework shall provide a web-based user interface accessible through supported modern web browsers. Separate interface views shall be provided for users, administrators, and human agents according to their authorized functions.

User Interface: - Registration and Login: Forms for creating an account and securely logging in. - Role Selection Screen: Displays the list of role assistants (e.g., HR Assistant, IT Support Assistant, Admissions Assistant) available to the user. - Conversation Interface: A chat-style interface with a microphone control for voice input, a text box for typed queries, and a transcript of the ongoing conversation with spoken playback of responses. - Conversation History Interface: Displays previously held conversations and their outcomes. - Escalation Notification: Displays a message when a query is escalated to a human agent and notifies the user when the agent responds.

Administrator Interface: - Administrator Login: Provides secure access to administrative functions. - Role Profile Configuration Interface: Allows creation and editing of role profiles, intents, sample utterances, and workflow mappings. - Integration Management Interface: Allows configuration of connectors to internal organizational systems. - Access Control Interface: Allows management of user, administrator, and human-agent roles and permissions. - Admin Dashboard: Provides an overview of assistant usage and key statistics. - Reports and Analytics Interface: Provides analytical reports on intents, escalation rates, and role-wise usage.

Human Agent Interface: - Escalation Queue Interface: Lists conversations escalated to human agents. - Conversation Handling Interface: Allows the agent to view conversation context and respond to the user.

General UI Guidelines: - The interface shall use a simple, clear, and consistent layout across all screens. - Navigation elements shall be placed consistently to allow users to move easily between the dashboard, role selection, conversation, history, and logout functions. - Standard controls such as Submit, Cancel, Speak, Send, Search, Filter, and Logout shall be used where applicable. - The system shall perform input validation and display clear, understandable error messages for invalid, incomplete, or unrecognized input. - The interface shall provide clear visual indication of microphone status

(listening/idle) and of the current conversation state. - The interface shall provide accessibility considerations, including readable text, clear labels, sufficient contrast, and keyboard-accessible controls, in addition to voice interaction. - Access to screens and functions shall be controlled according to the user's authorized role.

3.2 Software Interfaces

The Role-Specific AI Voice Assistant Framework shall interact with the following software components:

Backend Framework: Flask (or FastAPI) shall be used as the web application framework for implementing the server-side logic, handling requests from the client interface, orchestrating speech processing and intent recognition, executing workflows, and communicating with the database.

Database: MySQL shall be used for persistent storage of user details, role profiles, intents, workflow mappings, conversation history, escalation records, integration configuration, and analytics data.

Speech-to-Text / Text-to-Speech Engine: The system shall use a speech-to-text component to transcribe spoken user queries into text and a text-to-speech component to convert generated textual responses into spoken audio played back to the user.

NLU / Intent Recognition Engine: The system shall use a natural-language-understanding component to classify the recognized intent of a user query and extract relevant entities, using the intents configured for the currently selected role.

Internal Organizational System Integrations: The workflow automation engine shall communicate with configurable internal-tool connectors (for example, an HRMS API for leave balance queries, an ITSM API for raising support tickets, or a CRM/admissions API for inquiry status) using REST-based API calls. For this academic prototype, these integrations shall be represented using mock or sandbox services.

Authentication and Authorization: The system shall provide authentication for registration and login, and role-based access control shall restrict access to administrative and human-agent functions.

3.3 Communications Interfaces

- **Client-Server Communication:** Communication between the client browser and the Flask/FastAPI backend shall use HTTP/HTTPS, with HTTPS preferred for deployment to protect data in transit.
- **Voice Streaming:** Audio captured through the browser microphone shall be transmitted to the speech-to-text component using standard web audio APIs and HTTP(S)/WebSocket communication as appropriate.
- **Data Transfer:** User, role-profile, conversation, and integration data shall be exchanged between the client, backend, and database using standard web request/response mechanisms.
- **Notification Communication:** The system shall notify users regarding escalation and resolution events through the web interface.
- **Network Requirement:** A stable internet connection shall be required to access the application and to stream voice input/output.
- **Communication Security:** Communication containing user credentials, conversation content, and integration data shall be

protected using secure communication mechanisms such as HTTPS during deployment.

(Hardware interfaces will not be implemented for this project.)

4. Analysis Models

Use Case Diagram

Figure 4.1: Use case diagram of the Role-Specific AI Voice Assistant Framework

Actors: Employee (User), Organization Administrator, Human Agent.

Use cases include: Login/Authenticate, Select Role Assistant, Configure Role Profile, Speak/Type Query, Recognize Intent, Manage Integrations, Execute Workflow, Receive Voice Response, View Analytics Dashboard, Escalate to Human Agent, View Conversation History, Manage Role Access Control, Handle Escalated Conversation.

The use case diagram illustrates the interactions between the three primary actors – Employee (User), Organization Administrator, and Human Agent – and the major functions provided by the system. Employees primarily interact with role selection, query submission, workflow execution, response delivery, and escalation. Administrators configure role profiles, manage integrations and access control, and monitor usage through the analytics dashboard. Human agents handle escalated conversations.

Activity Diagram

Figure 4.2: Activity diagram of a role-specific conversation

Flow: Start → User logs in → Select role assistant (HR/IT/Admissions) → Speak or type query → Convert speech to text → Recognize intent & extract entities → Validate intent confidence → [High confidence] Execute mapped workflow/API call → Generate response text → Convert response to speech → Deliver response to user; [Low confidence] Escalate to human agent → Deliver response to user → Log conversation and analytics → End.

The activity diagram represents the workflow of a single conversation with the assistant, beginning with user login and role selection, followed by capturing the spoken or typed query, converting speech to text, recognizing intent, and validating confidence. High-confidence intents are routed to the workflow automation engine for execution and a spoken response; low-confidence intents are escalated to a human agent. The conversation concludes with logging of the interaction for analytics.

5. System Features

5.1 User Registration and Login

5.1.1 Description and Priority This feature allows employees, administrators, and human agents to create an account and securely log in to the framework. Authentication is required to access protected functions such as role assistant selection, conversation history, and administrative or agent functions. *Priority: High — Benefit: 9, Penalty: 9, Cost: 4, Risk: 2*

5.1.2 Stimulus/Response Sequences - User selects “Register” → System displays the registration form. - User enters the required registration details and submits the form → System validates the information. - Registration information is valid → System creates the account and displays a confirmation. - User enters valid login credentials and selects “Login” → System authenticates the user and redirects to the appropriate dashboard based on role. - User enters invalid login credentials → System rejects the login attempt and displays an appropriate error message. - An unauthenticated user attempts to access a protected function → System denies access and prompts the user to log in.

5.1.3 Functional Requirements

ID	Requirement
REQ-1.1	The system shall allow a new user to register by providing the required registration information.
REQ-1.2	The system shall validate the registration information before creating an account.
REQ-1.3	The system shall prevent registration using information already associated with an existing account and shall display an appropriate error message.
REQ-1.4	The system shall allow a registered user to log in using valid authentication credentials.
REQ-1.5	The system shall deny access when invalid login credentials are provided and shall display an appropriate error message.
REQ-1.6	The system shall restrict access to protected system functions to authenticated users.
REQ-1.7	The system shall provide access to system functions based on the authenticated user’s authorized role (User, Administrator, or Human Agent).

5.2 Role Assistant Selection

5.2.1 Description and Priority This feature allows a logged-in user to select which role-specific assistant (e.g., HR Assistant, IT Support Assistant, Admissions Assistant) they wish to converse with, based on the role profiles configured for the organization. *Priority: High — Benefit: 8, Penalty: 6, Cost: 3, Risk: 2*

5.2.2 Stimulus/Response Sequences - User logs in → System displays the list of available role assistants configured for the organization. - User selects a role assistant → System loads the intents and workflow mappings associated with that role and opens the conversation interface. - No role assistants are configured → System displays an appropriate message indicating that no assistants are currently available.

5.2.3 Functional Requirements

ID	Requirement
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REQ-2.1	The system shall display the list of role assistants currently configured and active for the organization.
REQ-2.2	The system shall allow a user to select a role assistant before starting a conversation.
REQ-2.3	The system shall load the intents, sample utterances, and workflow mappings associated with the selected role profile.
REQ-2.4	The system shall display an appropriate message when no role assistants are available for selection.

5.3 Voice and Text Query Capture

5.3.1 Description and Priority This feature allows a user to submit a query to the selected role assistant either by speaking through the device microphone or by typing text, and converts spoken input into text for downstream processing. *Priority: High — Benefit: 9, Penalty: 8, Cost: 6, Risk: 5*

5.3.2 Stimulus/Response Sequences - User selects the microphone control and speaks a query → System captures the audio and converts it to text using the speech-to-text engine. - User types a query into the text box and submits it → System accepts the typed text directly. - The captured audio is unclear or empty → System displays an appropriate message asking the user to repeat the query. - Speech-to-text conversion fails due to a technical error → System displays an error message and allows the user to retry or type the query instead.

5.3.3 Functional Requirements

ID	Requirement
REQ-3.1	The system shall allow a user to submit a query using voice input captured through the device microphone.
REQ-3.2	The system shall allow a user to submit a query by typing text as an alternative to voice input.
REQ-3.3	The system shall convert captured voice input into text using a speech-to-text engine.
REQ-3.4	The system shall display an appropriate message when voice input cannot be captured or converted successfully.
REQ-3.5	The system shall pass the resulting query text to the intent recognition engine for further processing.

5.4 Intent Recognition and Entity Extraction

5.4.1 Description and Priority This feature analyzes the text of a user query, using the intents configured for the currently selected role, to determine the user's intent and extract relevant entities (such

as dates, ticket categories, or employee identifiers) needed to execute the corresponding workflow. *Priority: High — Benefit: 9, Penalty: 8, Cost: 7, Risk: 6*

5.4.2 Stimulus/Response Sequences - A query is received in text form → System analyzes the text against the intents configured for the selected role assistant. - An intent is recognized with sufficient confidence → System extracts relevant entities from the query and passes the intent and entities to the workflow automation engine. - No intent is recognized with sufficient confidence → System flags the query for escalation to a human agent. - Required entities are missing from the query → System prompts the user for the missing information before proceeding.

5.4.3 Functional Requirements

ID	Requirement
REQ-4.1	The system shall analyze the text of each user query to determine its intent, using the intents configured for the currently selected role.
REQ-4.2	The system shall extract relevant entities from the user query required to execute the corresponding workflow.
REQ-4.3	The system shall compute a confidence score for the recognized intent.
REQ-4.4	The system shall flag a query for escalation when the intent confidence score falls below a configured threshold.
REQ-4.5	The system shall prompt the user for missing required entities before executing the mapped workflow.

5.5 Workflow Automation and Execution

5.5.1 Description and Priority This feature executes the workflow mapped to a recognized intent, which may include invoking an internal organizational system integration (e.g., checking leave balance in an HRMS, raising a ticket in an ITSM) and preparing the information needed to generate a response. *Priority: High — Benefit: 9, Penalty: 8, Cost: 7, Risk: 6*

5.5.2 Stimulus/Response Sequences - An intent and its extracted entities are received → System looks up the workflow mapped to that intent for the current role. - The mapped workflow requires an internal-tool integration call → System invokes the configured connector with the relevant entities. - The integration call succeeds → System receives the result and prepares it for response generation. - The integration call fails or times out → System logs the failure and prepares a fallback response, optionally escalating the query to a human agent.

5.5.3 Functional Requirements

ID	Requirement
REQ-5.1	The system shall map each recognized intent to a corresponding workflow

	definition for the current role.
REQ-5.2	The system shall execute the mapped workflow, including invoking configured internal-tool integrations where required.
REQ-5.3	The system shall pass the extracted entities as parameters to the invoked integration.
REQ-5.4	The system shall handle integration failures gracefully and shall log the failure for administrative review.
REQ-5.5	The system shall escalate the query to a human agent when the mapped workflow cannot be completed successfully.

5.6 Response Generation and Text-to-Speech

5.6.1 Description and Priority This feature generates a natural-language response based on the outcome of the executed workflow and converts it into spoken audio, in addition to displaying it as text in the conversation interface. *Priority: High — Benefit: 8, Penalty: 6, Cost: 5, Risk: 4*

5.6.2 Stimulus/Response Sequences - A workflow completes execution → System generates a natural-language response describing the outcome. - The generated response is ready → System converts the response text into speech using the text-to-speech engine and plays it back to the user. - Text-to-speech conversion fails → System displays the response as text only and logs the failure.

5.6.3 Functional Requirements

ID	Requirement
REQ-6.1	The system shall generate a natural-language response based on the result of the executed workflow.
REQ-6.2	The system shall convert the generated response into speech using a text-to-speech engine.
REQ-6.3	The system shall display the generated response as text in the conversation interface in addition to playing the spoken response.
REQ-6.4	The system shall display the response as text only when text-to-speech conversion is unavailable or fails.

5.7 Role Profile Configuration

5.7.1 Description and Priority This feature allows an organization administrator to create and maintain role profiles, each consisting of a role name, a set of intents with sample utterances, and the workflow mappings associated with each intent. *Priority: High — Benefit: 9, Penalty: 7, Cost: 7, Risk: 6*

5.7.2 Stimulus/Response Sequences - Administrator selects "Create Role Profile" → System displays a form to enter the role name and description. - Administrator adds an intent with sample utterances and a workflow mapping → System validates and stores the intent definition under the role profile. - Administrator submits the completed role profile → System validates the configuration and activates the role assistant for user selection. - Administrator edits an existing role profile → System updates the stored configuration and reflects the change for future conversations. - Required configuration fields are missing or invalid → System displays an appropriate error message.

5.7.3 Functional Requirements

ID	Requirement
REQ-7.1	The system shall allow an authorized administrator to create a new role profile with a name and description.
REQ-7.2	The system shall allow an administrator to define one or more intents, with sample utterances, under a role profile.
REQ-7.3	The system shall allow an administrator to map each intent to a corresponding workflow.
REQ-7.4	The system shall validate role-profile configuration before activating a role assistant.
REQ-7.5	The system shall allow an administrator to edit or deactivate an existing role profile.
REQ-7.6	The system shall make an activated role profile available for selection by users.